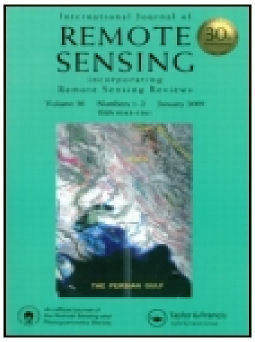

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Letter

Relating spectral shape to cyanobacterial blooms in the Laurentian Great Lakes

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A change in the spectral shape at 681 nm is used to distinguish blooms of cyanobacteria from blooms of other phytoplankton via MERIS satellite sensor imagery. During large cyanobacterial blooms, the spectral shape around 681 nm is not a positive quantity as scattering due to cyanobacteria overwhelms the fluorescence signal, thus creating a negative spectral shape. This relationship is consistent in both remotely sensed and *in situ* data.

1. Introduction

Blooms of the toxic cyanobacterium, *Microcystis aeruginosa*, have become common summer occurrences in certain areas of the Laurentian Great Lakes. These blooms in the Great Lakes have been accompanied by high toxin concentrations, which have caused much concern for water quality managers and the general public. Thus, the ability to accurately delineate high concentrations of *M. aeruginosa* is a public health priority in the Great Lakes.

The medium resolution imaging spectrometer (MERIS) has a higher spectral resolution in the red portion of the electromagnetic spectrum relative to other currently available ocean colour sensors (e.g. SeaWiFS and MODIS). As a result, MERIS has a greater potential than other sensors to discriminate blooms of cyanobacteria (Kutser 2004, Simis *et al*. 2005, Tomlinson *et al*. accepted).

Monitoring aquatic areas with remote sensing is advantageous as it allows for synoptic coverage. Ocean colour satellites such as SeaWiFS, MODIS and MERIS provide the temporal resolution needed to develop forecasts of conditions in a timely manner. One area that has experienced severe impacts from *M. aeruginosa* has been the western portion of Lake Erie. This area is in optically complex (case-2) water, where the optics of the water is governed by a combination of phytoplankton, sediment, and detrital and dissolved pigments. Traditional chlorophyll algorithms

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use a ratio of blue to green reflectance. Unfortunately, these bands are sensitive to high concentrations of detrital pigments that may interfere with chlorophyll detection. Algorithms that use bands within the red portion of the electromagnetic spectrum are not subject to this interference. The reflectance peak around 709 nm has been used in several other studies for quantitative chlorophyll assessment in case-2 waters (Gons 1999, Gitelson *et al*. 2000, Dall'Olmo and Gitelson 2005). Fluorescent line height (FLH), which is the spectral shape at 681 nm, has been used to detect chlorophyll. Gower *et al*. (1999) developed an index termed the maximum chlorophyll index, MCI, which is the spectral shape at 709 nm, to find harmful algal blooms. These applications indicate that spectral shape may provide a method for separating different blooms. The objective of this study was to determine if spectral shape in the red wavelengths can be used to differentiate cyanobacterial blooms from other algal blooms in the case-2 waters of the Great Lakes.

2. Methods

Standard level 2 MERIS satellite sensor imagery was obtained from the European Space Agency (Montagner 2001). The spectral shape at 681 nm was determined by:

$$\mathrm{SS}(\lambda)=\mathrm{nLw}(\lambda)-\mathrm{nLw}(\lambda^-)-\{\mathrm{nLw}(\lambda^+)-\mathrm{nLw}(\lambda^-)\}\frac{(\lambda-\lambda^-)}{(\lambda^+-\lambda^-)} \tag{1}$$

where SS is the spectral shape, nLw is the normalized water leaving radiance, λ is 681 nm, λ^+ is 709 nm and λ^- is 665 nm. A positive SS(681) is the same as FLH. This equation reduces to a form equivalent to the second derivative if the bands are evenly spaced. The results from this analysis indicate that the SS(681) was negative. This is caused by the nLw(681) (the fluorescence band) falling below the baseline during known blooms of cyanobacteria. The baseline is defined as a straight line drawn between the nLw at 665 nm (λ^-) and the nLw at 709 nm (λ^+).

Field radiometry was obtained from a bloom of *M. aeruginosa* in Bear Lake, Michigan, USA (43.25° N, −86.28° W) on 1 August 2006 (figure 1) and from a bloom of other phytoplankton (not cyanobacteria) from Saginaw Bay (43.94° N, −83.33° W) on 28 August 2005. Above-water radiances were measured using a handheld spectrometer that collects surface radiance from 350 to 900 nm at 1.04 nm spectral resolution with a 2.5 nm half bandwidth. All water spectra were taken about 1 m above the surface of the water. The reflectance from sky view data was calculated, incorporating a calibrated spectralon reference plate (Mueller *et al*. 2003). All radiance measurements were carried out in triplicate, averaged and converted to remote sensing reflectance (Rrs). Each Rrs value was multiplied by its respective solar constant to determine nLw. The results were used in equation (1) to calculate SS(681).

3. Results

A time-series from the summer of 2003 was processed and examined for western Lake Erie. The western portion of the lake had a large bloom of cyanobacteria present beginning in mid-August (Rinta-Kanto *et al*. 2005). This feature was present in a Landsat-5 image captured on 19 August 2003 (figure 2). The time-series of the spectral shape at 681 nm is seen in figure 3. The positive SS(681) values are shown in black and the negative SS(681) values are shown in colour. Warmer colours are indicative of a highly negative SS(681). The shape and extent of the cyanobacterial

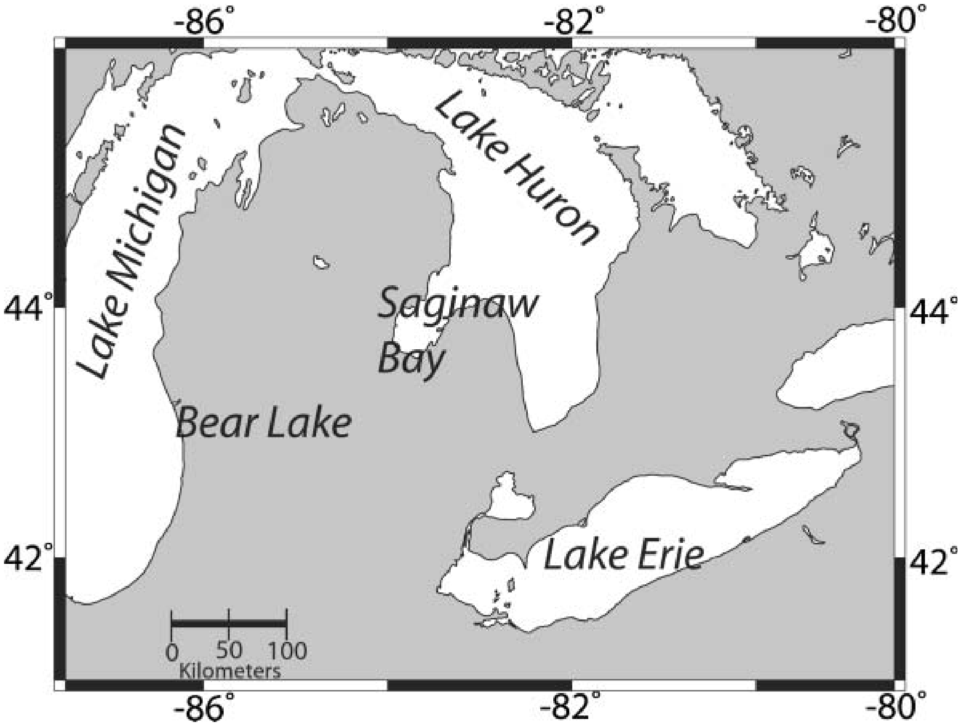

Figure 1. A map showing the study area.

bloom are shown to have a negative spectral shape at 681 nm throughout the time-series. The feature seen by the negative SS(681) matches very well with the blue-green feature present in the Landsat scene in figure 2. The image from 21 August 2003 (figure 3), the clearest image in the time-series, was used to further illustrate this premise. A polygon was drawn around the (colour) feature shown in figure 3 from the derived SS(681), and all available nLw values were extracted from the polygon and spatially averaged. The number of pixels (*n*) used for this analysis was 346 with the resultant spectra plotted in figure 4. The same polygon was then overlaid onto an image captured from 17 August 2004, nearly the anniversary date of the first image. Once again, all available nLw values were extracted and spatially averaged ($n=346$). There was a small cyanobacterial bloom in 2004, but it was concentrated further to the east (Rinta-Kanto *et al.* 2005), and the bloom was not represented in the polygon used. Hence we used this extracted data to represent non-bloom conditions. The bloom spectra shows a negative SS(681), while the non-bloom spectra show a positive SS(681).

The above findings are consistent with results collected from field data. The SS(681) showed a negative shape in all field and satellite samples that coincided with the presence of a bloom of cyanobacteria. Two examples of *in situ* spectra are shown in figure 5. In the cyanobacterial bloom, the nLw(681) falls below the baseline, hence a negative SS(681). In the case without a cyanobacterial bloom (non-bloom conditions), the nLw(681) falls above the baseline, leading to a positive SS(681). These results are consistent with the MERIS satellite sensor imagery results.

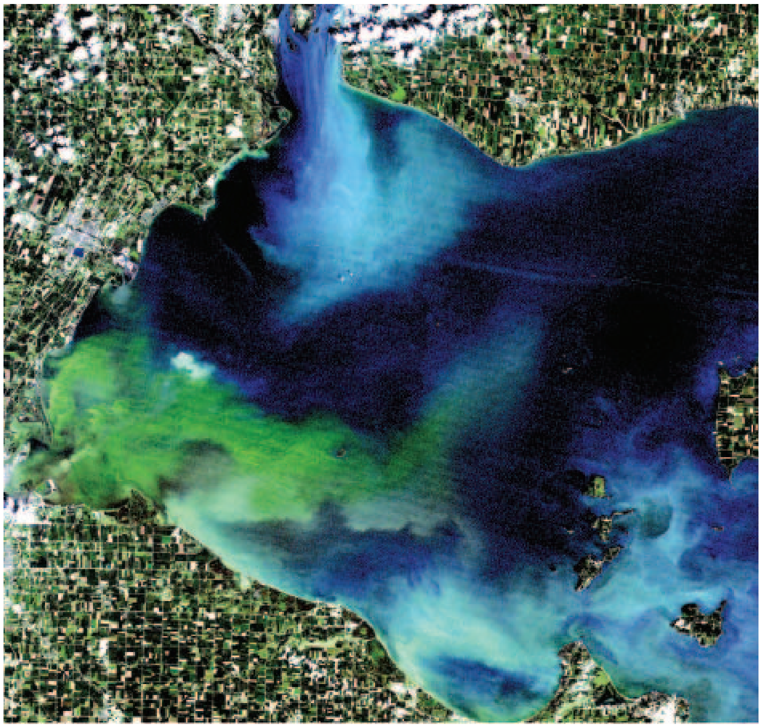

Figure 2. A true colour Landsat image from 19 August 2003 showing a large bloom of cyanobacteria (Image from Landsat 5 server courtesy of OhioView).

4. Discussion

Our results suggest that a negative SS(681) occurs in the presence of large concentrations of *M. aeruginosa* and can, therefore, be used to separate them from other types of phytoplankton. This phenomenon may occur for several reasons. Seppälä *et al*. (2007) have noted that cyanobacteria have a much lower fluorescence signal than eukaryotic phytoplankton. In addition, Ganf *et al*. (1989) estimated that over 80% of light scattering in natural populations of *M. aeruginosa* was from the presence of gas vacuoles, generally making it a much stronger scatterer of light than most phytoplankton species. Therefore, within a growing bloom population, fluorescence appears to be overwhelmed by the scattering properties of *M. aeruginosa*.

Typically, high concentrations of phytoplankton should fluoresce, leading to a reflectance peak around 681 nm (positive SS(681)), as seen in the non-bloom examples (see grey lines in figures 4 and 5). When phytoplankton and scatterers are present in large concentrations, the strong scattering greatly exceeds fluorescence, causing a peak at 700–710 nm, just to the right of 681 nm (Gitelson 1992). Even with the peak reflectance at 709 nm, weakly and strongly fluorescing phytoplankton appear to show differences at 681 nm. Algae that strongly fluoresce may produce a shoulder on the peak at 681 nm, namely positive SS(681). For high concentrations of *M. aeruginosa*, weak fluorescence may produce a 'sag' at 681 nm, causing negative SS(681). The data on chlorophyll concentration suggest that the method may apply to concentrations as low as 5 $\mu g\,L^{-1}$. Further evaluation of this hypothesis is needed,

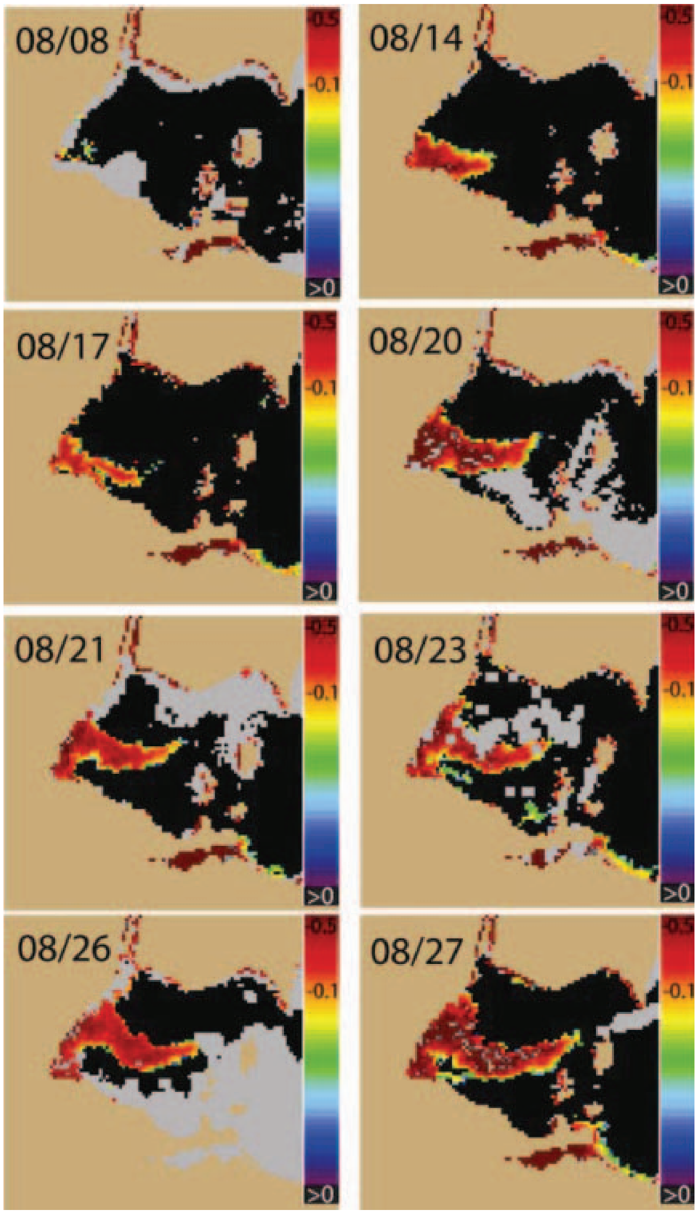

Figure 3. A time-series of the spectral shape at 681 nm derived from MERIS imagery. Areas with a negative SS(681) are highlighted in colour, no data are shown in grey. Areas with a positive SS(681) are shown in black. The areas with a negative SS(681) are consistent with the location of the cyanobacterial bloom. Note how the shape of the negative SS(681) area closely follows the blue-green feature seen in figure 2.

with other blooms and in water with non-algal scattering, but initial results appear promising.

Scattering properties of individual cyanobacterial species and the surrounding water must be taken into consideration when applying spectral shape algorithms. For instance, some areas of the bloom in figure 2 exhibited surface scum. However, some cyanobacteria do not form surface scums, such as *Cylindrospermopsis* spp.

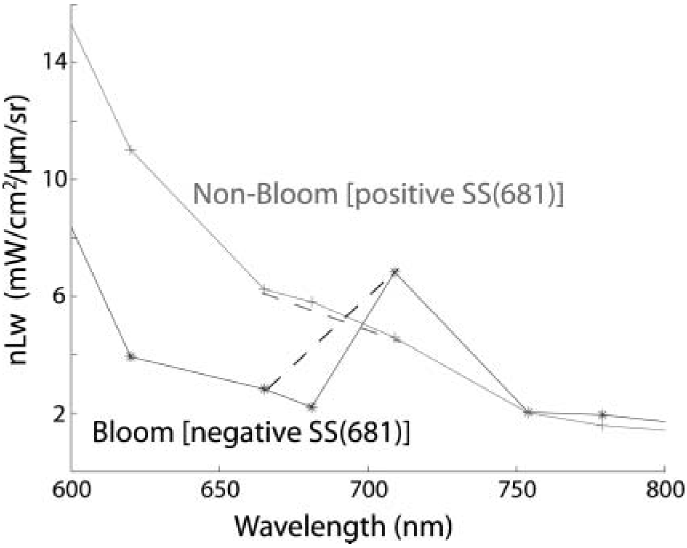

Figure 4. Spectra from two clear MERIS images in western Lake Erie. The black line is derived from data captured on 21 August 2003 and illustrates cyanobacterial bloom conditions. The grey line is derived from data captured on 17 August 2004 and illustrates non-bloom conditions. The bloom image shows a negative SS(681), where the non-bloom image shows a positive SS(681).

and, therefore, may not exhibit the same optical properties from 665 to 709 nm. The peak reflectance at around 709 nm can occur in assemblages of other species of phytoplankton, if the phytoplankton occurs in a scattering medium (i.e. sediment-laden waters) (Gitelson *et al*. 2000). During the study period there was a relatively

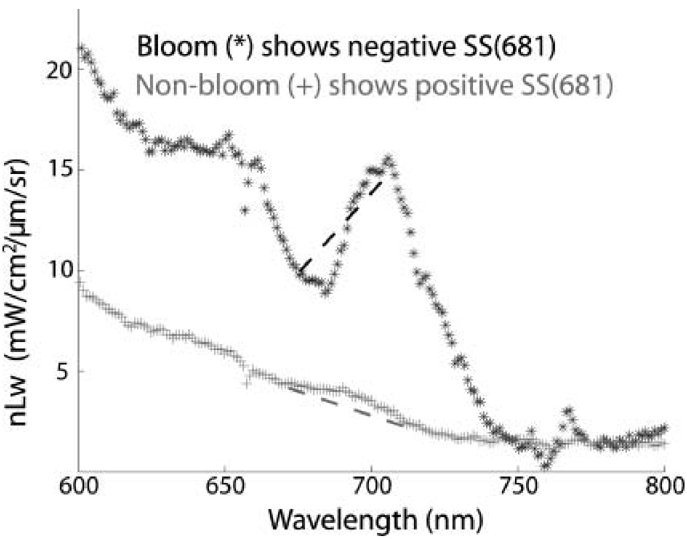

Figure 5. *In situ* reflectance signatures from different locations. The black line (*) shows the spectra from a bloom in Bear Lake and exhibits a negative SS(681). The dashed black line is the baseline. The grey line (+) shows the spectra from Saginaw Bay during a time without a cyanobacterial bloom and shows a negative SS(681). The dashed grey line is the baseline.

low wind stress, indicating that there most likely were low concentrations of suspended sediment in the water column. Cyanobacterial blooms generally occur during a stable, stratified water column and are not likely to occur during periods of high winds and resuspension (Paerl 1988). During strong mixing conditions, however, the scattering properties of *M. aeruginosa* may diminish when colonies are mixed to depth (Ganf *et al*. 1989), which could alter the results.

This paper follows on analyses of the red bands for chlorophyll and bloom detection (Gons 1999, Gower 1999, Gitelson *et al*. 2000). It shows that the use of spectral shape in the red bands may offer additional capabilities in differentiating blooms of cyanobacteria from blooms of other types of phytoplankton.

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References

Dall'Olmo, G. and Gitelson, A.A., 2005, Effect of bio-optical parameter variability on the remote estimation of chlorophyll-a concentration in turbid productive waters: experimental results. *Applied Optics*, **44**, pp. 412–422.

Ganf, G.G., Oliver, R.L. and Walsby, A.E., 1989, Optical properties of gas-vacuolate cells and colonies of *Microcystis* in relation to light attenuation in a turbid, stratified reservoir (Mount Bold Reservoir, South Australia). *Australian Journal of Freshwater Research*, **40**, pp. 595–611.

Gitelson, A., 1992, The peak near 700 nm on radiance spectra of algae and water: relationships of its magnitude and position with chlorophyll concentration. *International Journal of Remote Sensing*, **13**, pp. 3367–3373.

Gitelson, A.A., Yacobi, Y.Z., Schalles, J.F., Rundquist, D.C., Han, L., Stark, R. and Etzion, D., 2000, Remote estimation of phytoplankton density in productive waters. *Archives for Hydrobiolia – Special Issues in Advancements in Limnology*, **55**, pp. 121–136.

Gons, H.J., 1999, Optical teledetection of chlorophyll a in turbid inland waters. *Environmental Science and Technology*, **33**, pp. 1127–1132.

Gower, J.F.R., Doerffer, R. and Borstad, G.A., 1999, Interpretation of the 685 nm peak in water-leaving radiance spectra in terms of fluorescence, absorption, and scattering, and its observation by MERIS. *International Journal of Remote Sensing*, **20**, pp. 1771–1786.

Kutser, T., 2004, Quantitative detection of chlorophyll in cyanobacterial blooms by satellite remote sensing. *Limnology and Oceanography*, **49**, pp. 2179–2189.

Montagner, F., 2001, Reference model for MERIS level 2 processing. European Space Agency, document PO-TN-MEL-GS-0026.

Mueller, J.L., Davis, C., Arnone, R., Frouin, R., Carder, K., Lee, Z.P., Steward, R.G., Hooker, S., Mobley, C.D. and McLean, S., 2003, Above-water radiance and remote sensing reflectance measurement and analysis protocols. In *Ocean Optics Protocols for Satellite Ocean Color Sensor Validation, Revision 4, Volume III: Radiometric Measurements and Data Analysis Protocols*, J.L. Mueller, G.S. Fargion and C.R. McClain (Eds). NASA/TM-2003-211621/Rev4-Vol. III, NASA Goddard Space Flight Center, Greenbelt, Maryland, January 2003, pp. 21–31.

Paerl, H.W., 1988, Nuisance phytoplankton blooms in coastal, estuarine, and inland waters. *Limnology and Oceanography*, **33**, pp. 823–847.

Rinta-Kanto, J.M., Ouellette, A.J.A., Boyer, G.L., Twiss, M.R., Bridgeman, T.B. and Wilhelm, S.W., 2005, Quantification of toxic *Microcystis* spp. during the 2003 and 2004 blooms in western Lake Erie using quantitative real-time PCR. *Environmental Science and Technology*, **39**, pp. 4198–4205.

Seppälä, J., Ylöstalo, P., Kaitala, S., Hällfors, S., Raateoja, M. and Maunula, P., 2007, Ship-of-opportunity based phycocyanin fluorescence monitoring of the filamentous cyanobacteria bloom dynamics in the Baltic Sea. *Estuarine Coastal and Shelf Science*, **73**, pp. 489–500.

Simis, S.G.H., Peters, S.W.M. and Gons, H.J., 2005, Remote sensing of the cyanobacterial pigment phycocyanin in turbid inland water. *Limnology and Oceanography*, **50**, pp. 237–245.

Tomlinson, M.C., Wynne, T.T., Stumpf, R.P., Dyble, J., Fahnenstiel, G.L. and Tester, P.A., Accepted, Using remote sensing to aid in the detection and monitoring of *Microcystis aeruginosa* in western Lake Erie and Saginaw Bay, USA. In *Proceedings of the XII International Conference on Harmful Algae*.